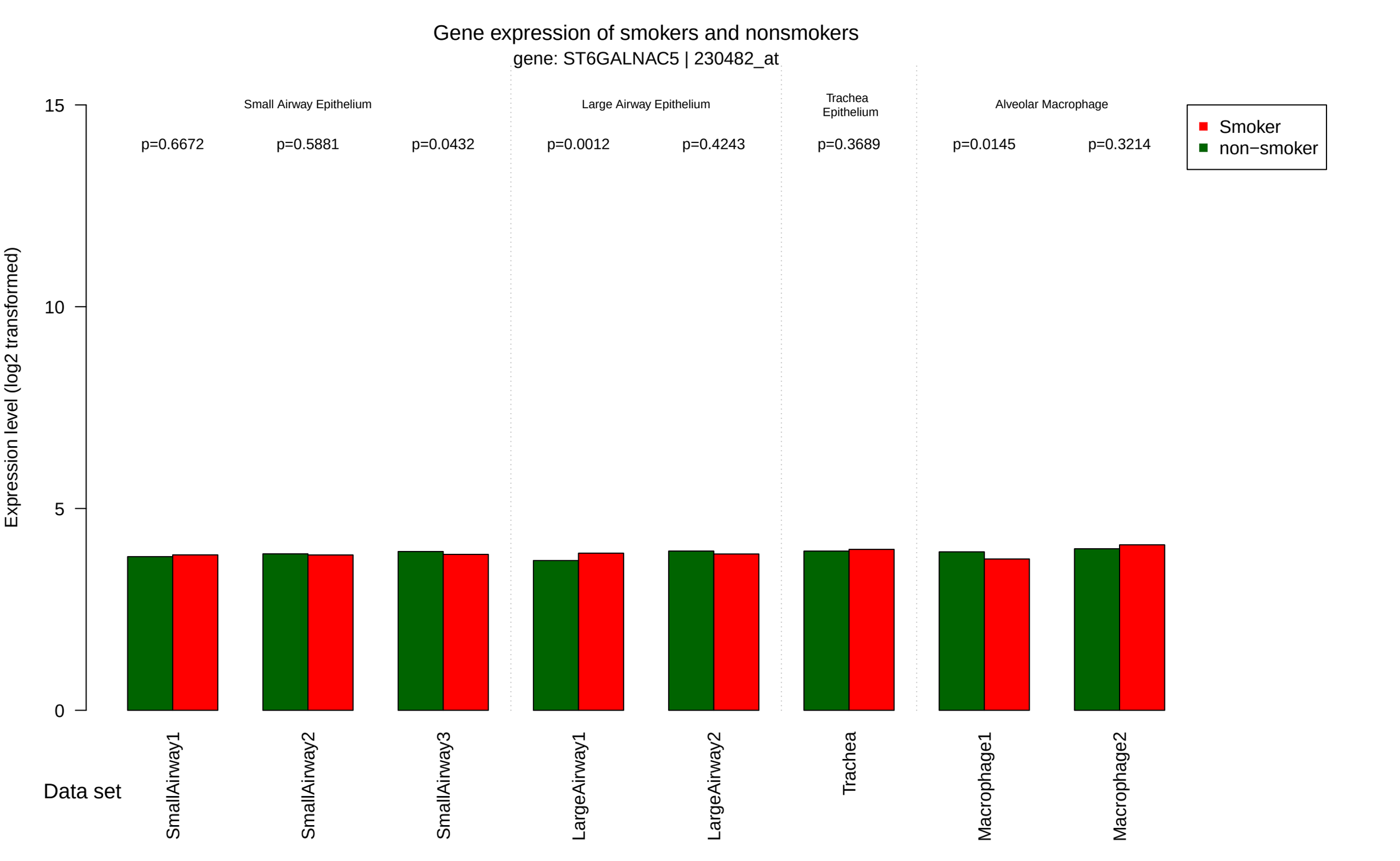
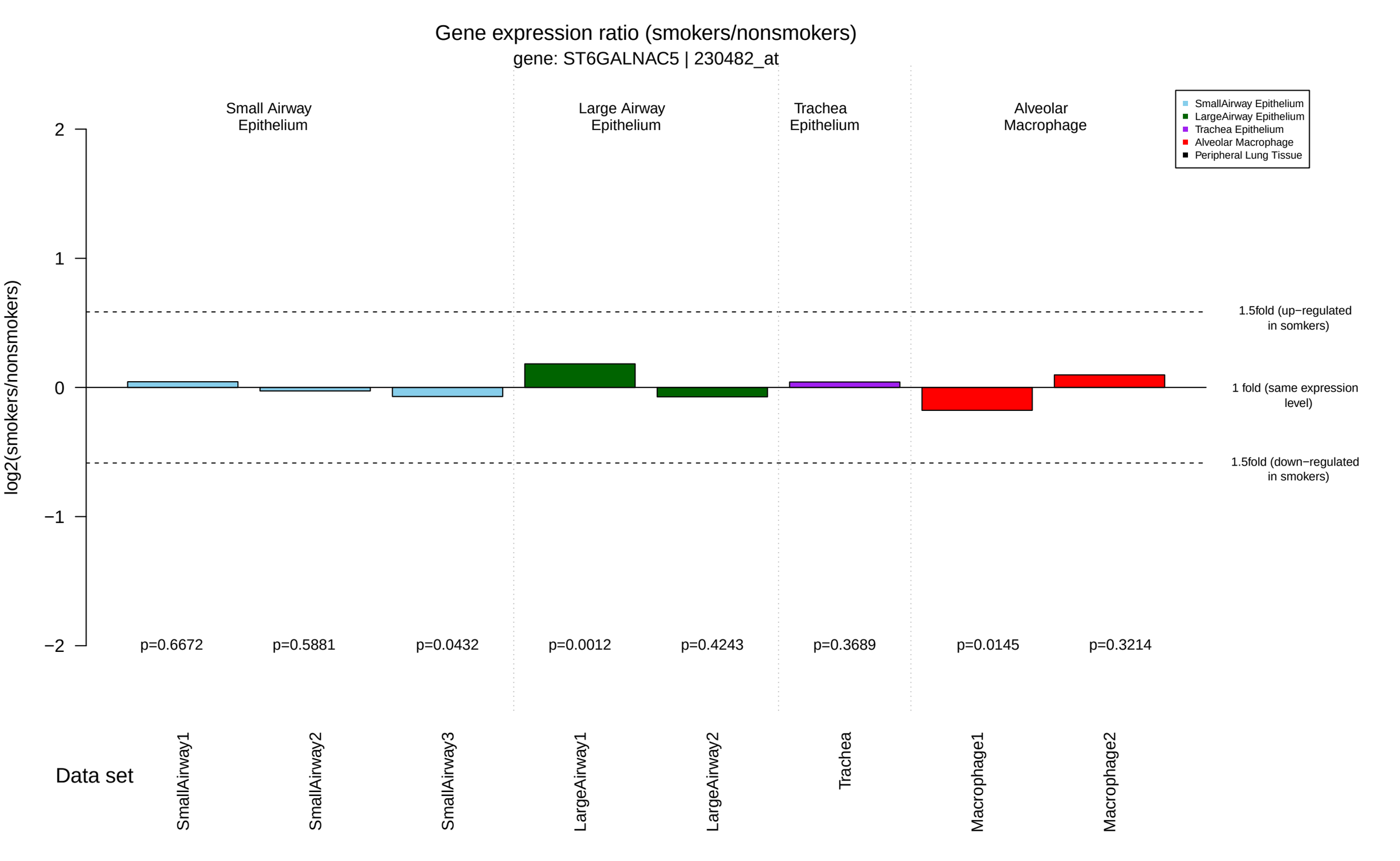
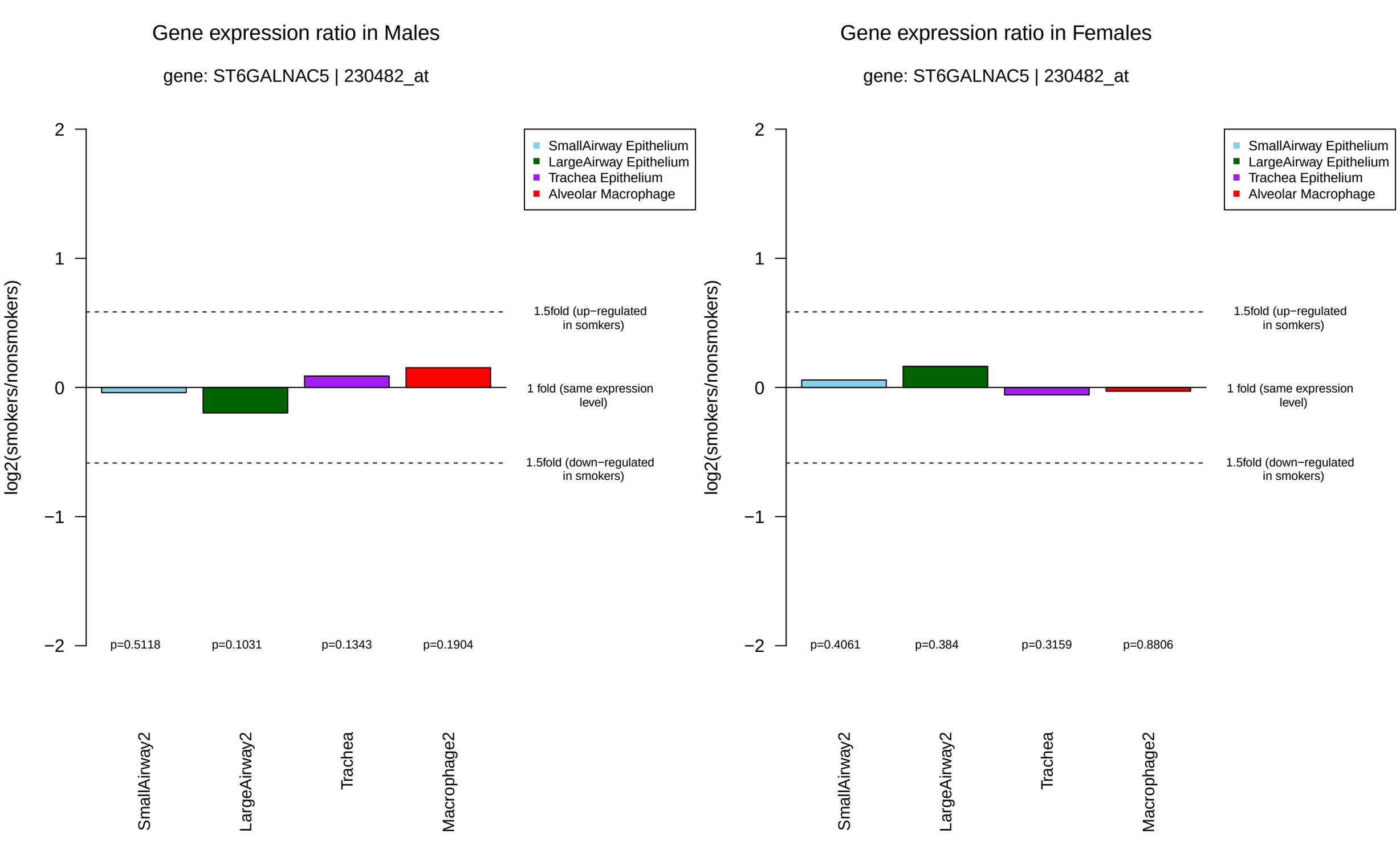
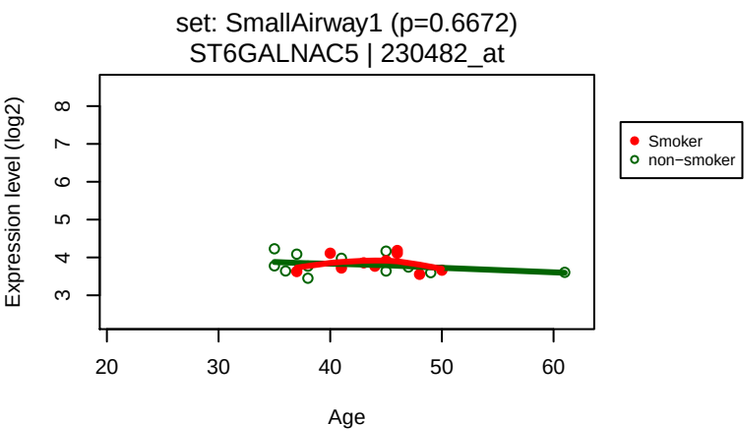
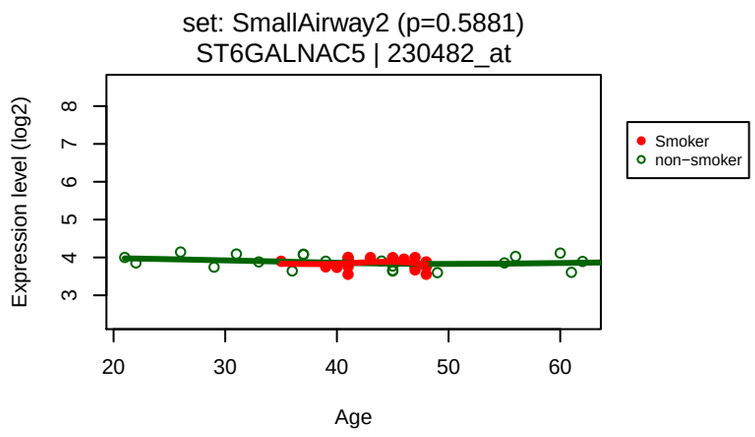
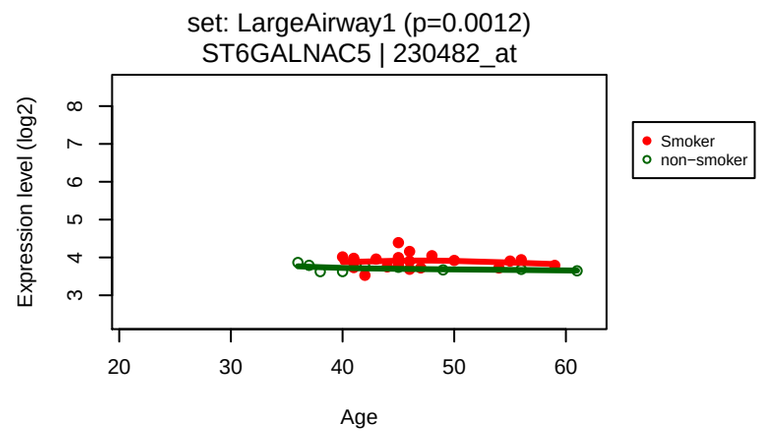

Gene expression of smokers and nonsmokers
gene: ST6GALNAC5 | 230482_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6672
p=0.5881
p=0.0432
p=0.0012
p=0.4243
p=0.3689
p=0.0145
p=0.3214
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

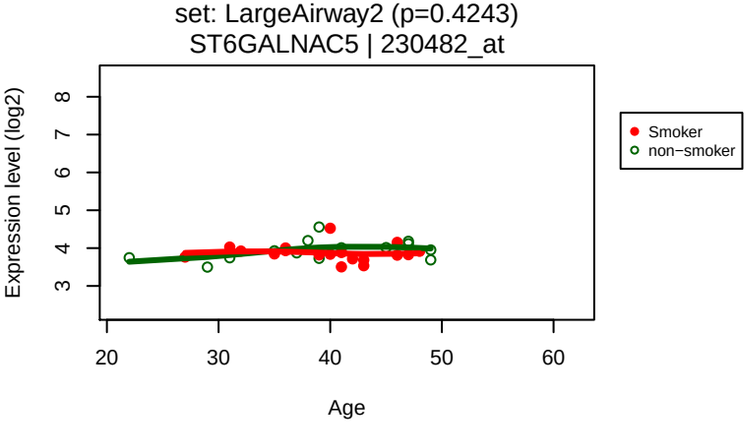

Gene expression ratio (smokers/nonsmokers)
gene: ST6GALNAC5 | 230482_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6672
p=0.5881
p=0.0432
p=0.0012
p=0.4243
p=0.3689
p=0.0145
p=0.3214
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

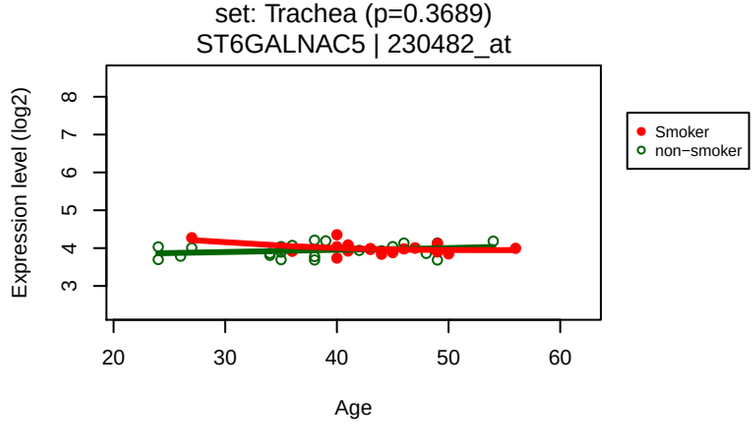

Gene expression ratio in Males
gene: ST6GALNAC5 | 230482_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5118
p=0.1031
p=0.1343
p=0.1904
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST6GALNAC5 | 230482_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4061
p=0.384
p=0.3159
p=0.8806
SmallAirway2
LargeAirway2
Trachea
Macrophage2

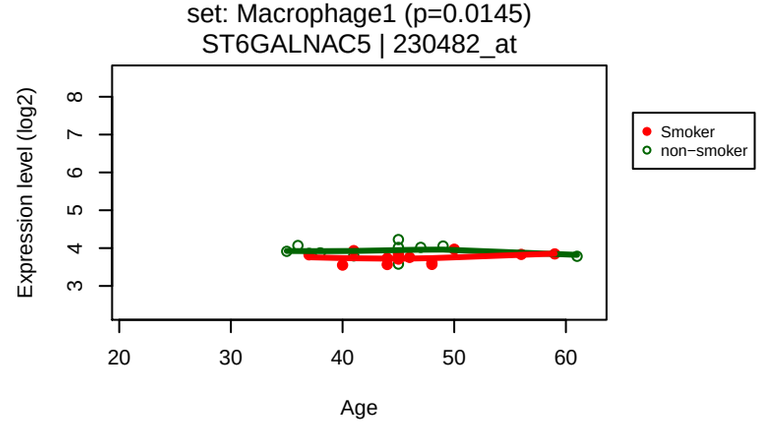

set: SmallAirway1 (p=0.6672)
ST6GALNAC5 | 230482_at
Expression level (log2)
Age
Smoker
non−smoker

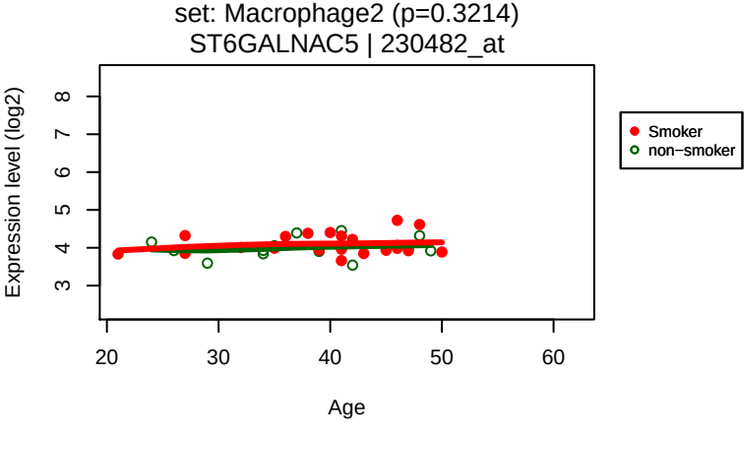

set: SmallAirway2 (p=0.5881)
ST6GALNAC5 | 230482_at
Expression level (log2)
Age
Smoker
non−smoker

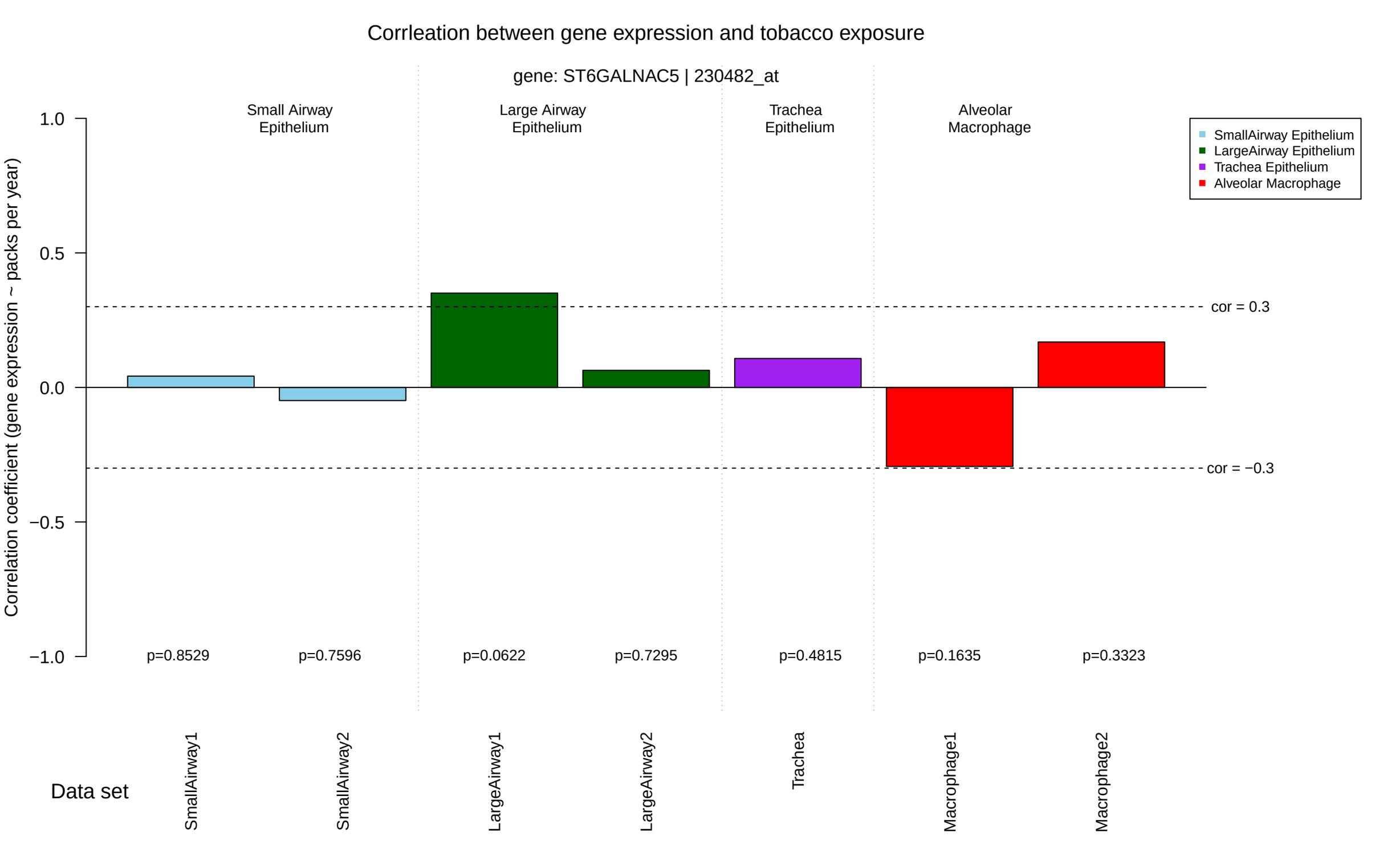

set: LargeAirway1 (p=0.0012)
ST6GALNAC5 | 230482_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4243)
ST6GALNAC5 | 230482_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3689)
ST6GALNAC5 | 230482_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0145)
ST6GALNAC5 | 230482_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3214)
ST6GALNAC5 | 230482_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST6GALNAC5 | 230482_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8529
p=0.7596
p=0.0622
p=0.7295
p=0.4815
p=0.1635
p=0.3323
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2